Whose Contribution Counts?

AI Mediation and Collective Authorship in Digital Improvisation

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PhD Dissertation Proposal (research-creation option)

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Summary

This research asks what happens to people's creative contributions when a generative AI system turns them into part of a shared artwork. In small-group improvisation workshops, participants will type short phrases that a language model turns into light, form and sound in a shared projected scene. The study follows how the group recognises, accepts, changes or refuses these transformations, and who comes to be credited and held responsible for the result. It contributes an explanation of when a system's record of who contributed what matches, and when it fails to match, how collaborators themselves understand authorship.

1 Introduction

In collaborative creative tools, a generative model does more than produce content: it *reinterprets* what people contribute. Someone types a phrase, and a moment later it has become a sound or a moving form in a shared scene. The group then has to decide what that transformation means, and whether it still belongs to the person who offered it. When AI translates contributions into shared audiovisual material, its outputs may become more visible or influential than the contributions that prompted them, a possibility to investigate rather than an established effect. This is a communication problem: Hancock et al. (2020) define AI-mediated communication as "interpersonal communication in which an intelligent agent operates on behalf of a communicator by modifying, augmenting, or generating messages to accomplish communication goals." Here the "message" is an artistic offer, and its modification becomes part of a collective artwork. This raises a cultural question about whose interpretations gain authority, whose contributions remain recognisable, and how disagreement is negotiated in collective creative practice. Creative platforms increasingly record who supplied what; this project asks whether such records settle questions of credit.

Improvisation is a good setting for the question. Improvisers already have conventions for listening to, building on and refusing one another's offers, and improvisation has been argued to be a research method grounded in risk, active listening and collaborative response (Caines, 2020). A system that transforms and labels offers enters those conventions, and may quietly change them. The project uses the program's research-creation option: I will build a shared audiovisual environment for improvisation and study the works made with it.

2 Objectives

2.1 Research problem and questions

The problem is the relationship between technical attribution, meaning the system's record of who supplied what, and social recognition, meaning how collaborators acknowledge, credit and hold one another responsible. Three working definitions organise the study. **Recognition**: a contribution is noticed or acknowledged as meaningful. **Creative authority**: a person's interpretation or decision carries influence over what happens next. **Authorship**: participants associate ownership, credit or responsibility with the work, as felt ownership, public attribution and creative responsibility. Legal ownership and copyright are outside the scope of this study. These can come apart: a contributor may be credited while the group reshapes the material regardless of their view (credit without influence), and a facilitator, interpreter or designer may shape the work uncredited (influence without credit). The definitions are sensitising concepts, not fixed categories. Participation includes listening, facilitation, silence and refusal as well as submitting material.

Central question. In collaborative digital improvisation, how do participants, facilitators and a generative-AI interface shape whether a person's contribution is recognised, carries creative authority, and is credited in the shared work?

RQ1. *Recognition.* How do participants notice, interpret and respond to differences between what they offered and how it was transformed?

RQ2. *Authority.* Through which practices, including acceptance, revision, refusal, facilitation and silence, do contributors and groups gain or lose influence over transformed material, and how do interface permissions shape those practices?

RQ3. *Authorship.* When and why do the system's technical attribution and participants' accounts of ownership, credit and responsibility diverge?

2.2 Theoretical perspective

The primary framework is Lucy Suchman's study of human–machine reconfiguration (Suchman, 2007), which examines how agencies are figured at the interface and how humans and machines come to be treated as similar or different in practice. On this view, whether a transformed element belongs to its contributor, to the group or to the system is worked out in interaction. Suppose the system records that an amber pulse came from one participant's phrase. The pulse also depends on the model's reading, on the asset library I built and on the approval rule in force. When the contributor refuses a request to brighten it, they enact an authority the interface granted, and the group can accept or contest it. The question shifts from "who really made the pulse?" to how agency is distributed and attributed in that moment. Three questions guide the analysis: how participants attribute agency and authorship in talk and action; how interface rules, model defaults, facilitation and my design decisions make some actions easy and others difficult; and who is held responsible when a transformation misrepresents someone. Improvisation studies adds a limited lens on how offers are taken up, ignored or refused (Caines, 2020). These concepts direct attention without fixing the findings.

2.3 Expected contribution and originality

The study investigates whether technical attribution and social recognition can diverge in collective creative work. A provenance log may record who supplied a phrase while collaborators treat the transformed material as another person's, the group's, or nobody's. The intended contribution is an explanation of how contributors gain or lose authority over transformed material, and of how interface permissions, facilitation and group conventions shape that process. It does not offer a general claim that more individual control produces fairer collaboration. The evidence may help explain credit without influence, influence without credit, how accepted material becomes collective, and when provenance fails to resolve authorship disputes; these are possibilities, not predicted findings. If AI proves unnecessary in some arrangements, the contribution would lie in explaining when and why.

Originality lies first in the object of study: the closest studies mostly position AI as a partner, performer or group resource that people create *with*, whereas this project studies AI as a medium that people create *through*, a translator between collaborators. It also lies in the combination examined. Prior work has studied group negotiation around a shared AI (Xu et al., 2026), improvisation as a basis for AI partners and performers (Trajkova et al., 2024; Mathewson and Mirowski, 2017), and observers' attribution of credit for AI art (Epstein et al., 2020). This project follows identifiable contributions through AI-mediated transformation in *repeated* collective improvisation, with a

human-mediated contrast and attention to facilitation, and compares technical attribution with socially negotiated recognition. The resulting concepts may help interpret other platforms that rely on attribution records, without establishing how those platforms behave.

3 Methods and sources

3.1 Research methods

The project follows the program's research-creation option, a creative work with a 150–200-page written dissertation (Joint Graduate Program in Communication & Culture, 2021). It is closest to what Chapman and Sawchuk (2012) call "research-from-creation": making the work generates material for understanding how participants engage with it.

The creative work. The creative component will comprise the shared audiovisual environment and the improvised works produced through it. Developing and revising its expressive possibilities, interaction rules and workshop scores is part of the inquiry into how creative authority is distributed. Participants build and perform short projected improvisations of light, form and sound, then discuss them; each rule the environment embodies is a proposition about collective authorship that they can test, resist or rewrite. In an illustrative session, four improvisers receive a score: build the weather of a city none of them has visited. One types "the bus that never comes." The system proposes a slow amber pulse at the edge of the space with a low drone, with the generated interpretation "waiting as a distant repeating light." The contributor accepts it. Later another participant wants the pulse brighter; the contributor declines, and the facilitator asks the group how to proceed. The workshop scores are a creative output in their own right: a revised set of improvisation scores for AI-mediated group work, credited to the groups that shaped them with their consent.

Ownership and reuse. Recording origins and giving contributors approval rights could introduce individual ownership where improvisers treat accepted offers as shared. The scoping interviews and co-design described below address this directly, and one option to test is a way to "release" accepted material to the group.

The system. A participant submits a short text phrase. An existing language model returns one proposal drawn from a curated library of forms, light, colour, particle "weather", sound and text, with a one-line generated interpretation that is model output, not an explanation of why the model produced the transformation. Contributors or groups can accept, edit, reject, request another proposal or undo, and a manual path to the same library is always open. Generative AI is included because machine-mediated reinterpretation is the object of study. The project does not presume

that AI improves improvisation. The model does not decide whether anything enters the work, who is credited or what is removed, and makes no inferences about participants' emotions or behaviour. Unity applies only permitted, bounded actions; model output is checked against them and never run as code. No model is trained or fine-tuned.

Phases.

1. *Scoping interviews* with improvisers and facilitators about how offers, credit, reuse and refusal already work, including whether accepted material becomes the group's and who decides that.
2. *Co-design* with a few practitioners to set the input form, the limits of transformation, the starting approval and attribution rules, and accessibility provisions. Co-design will examine whether contributor approval and visible attribution support practitioners' conventions or introduce individualised ownership assumptions they do not share. These practitioners will not join the workshop groups.
3. *Workshops.* Provisionally, three groups of three to five adult practitioners each attend four workshops of approximately 90–120 minutes. The first uses only the manual path, and the group writes down its own decision procedure. Each later workshop uses one arrangement, in a different order for each group: *contributor-led* (AI proposes, contributor approves); *group-led* (AI proposes, the group applies its procedure); *human-mediated* (a participant in a rotating interpreter role proposes with the same assets, contributor approves).
4. *Reflection interviews* with each participant, reviewing video of their own sessions.

Each group meets each arrangement once. What repeats is the group's evolving collaboration, not trials within an arrangement. The arrangements are situated contrasts whose effects cannot be separated from order, prior relationships and emerging norms. I treat these as contextual evidence, and the study will not isolate a causal effect of AI. During a session, participants may negotiate how rules apply, for instance by releasing material to the group or overruling an approval. Substantial system changes are made only between sessions, and each rule version, its reason and its consequences are recorded. A changed arrangement is not treated as equivalent to its earlier version; the change is analysed as evidence of negotiated authority.

Analysis. I will construct multimodal accounts of selected episodes, linking video of participants' words and actions with screen recordings and the system's log. An episode runs from a contribution, through its transformation and the group's response, to the next creative action. Selection includes routine acceptance and non-response as well as explicit disagreement. Selection decisions are documented, and episodes are compared across groups, arrangements and stages of collaboration, including cases that challenge emerging interpretations. The episode analysis adapts the video microethnography McArthur (2015) used to study interface interactions. Reflection interviews are

analysed through reflexive thematic analysis (Braun and Clarke, 2019), whose themes contextualise episodes without settling their meaning. System records show what happened in the software, not a "true" account of intention. Discrepant cases revise an emerging explanation or limit its scope. I will keep an analytic trail rather than seek inter-rater agreement, and a design journal records how my choices as designer and facilitator shape what participants can do.

Ethics. Consent will be specific, ongoing and revisable (Im et al., 2021; Strengers et al., 2021), given separately for recording, AI processing, use of recordings in the examination, and public dissemination; public sharing is optional. Provenance will support identifying and, where feasible, removing attributable contributions. Participants will agree on withdrawal procedures and deadlines beforehand, including the limits of removing contributions from jointly developed or publicly disseminated works. Participants are paid and credit is agreed with them; data are minimised, pseudonymised and kept for stated periods. The program's ethics form will be submitted with this proposal. Full participation without AI is a design commitment the workshops will test, through reduced motion, resizable text, alternatives to sound-only information, facilitator-operated controls and the manual path. Participation alone does not make a project equitable (Birhane et al., 2022).

3.2 Sources

The primary sources are generated by the study. They comprise interview recordings and transcripts from scoping and reflection interviews; video and screen recordings of workshops; the system's log of contributions, proposals and decisions; written group procedures and rule versions; the design journal; and the improvised works themselves. The creative component will be documented for examination through the executable environment, workshop scores, selected consented recordings, and an annotated account of design revisions. Secondary sources are the scholarship discussed in Section 4.

3.3 Anticipated problems of access

- *Participants and setting.* A possible primary context is a connection to Dr. Rebecca Caines's SSHRC Insight project, *Interfacing, Playtesting, Resetting: Reframing Technology Use Through Improvised Creative Technologies* (Goodfellow Craig, 2026). Only its title and funding are public, and any relationship would be determined with her. Otherwise I will recruit through open calls to Toronto improvisation, theatre and sound communities and to AMPD students, excluding anyone I teach. As in Chalk (2022), recruitment will be iterative, checking whose practices are missing after the first group and reporting remaining gaps.

- *Communities.* I will not work with disability-focused or Indigenous communities without established partnerships, community governance and relevant expertise.
- *AI services.* I will first test a local open-weight model to avoid third-party processing. If it is inadequate, I will use only an institutionally approved hosted service that receives only the phrase, subject to ethics review.
- *Retention.* If a group cannot complete all four workshops, its completed sessions remain usable as a smaller case, and recruitment of a replacement group will be considered.

4 Relationship to existing literature and practice

Stream. The project contributes to the Technology in Practice stream, with Media & Culture as a minor, at the meeting point of human–machine communication, participatory cultural production and research-creation.

Dominant perspectives. Human–AI co-creation research in HCI typically models how people and systems share control. Recent work extends this to groups: Xu et al. (2026) observed six student groups of three working around one shared ChatGPT interface in a single session and found AI output becoming a negotiable part of group decisions. They note that their analysis gave less attention to trust and accountability. Work on AI and improvisation positions AI as a partner or performer. Trajkova et al. (2024) studied dancers' partnered improvisation to design an AI partner whose recommendations had not yet been tested. Mathewson and Mirowski (2017) report on live performance with chatbot improvisors. Studies of credit, such as Epstein et al. (2020), show through vignette experiments that framing AI as an agent shifts the responsibility observers assign to people. Contestability has mainly been developed for high-stakes algorithmic decisions (Lyons et al., 2021). In communication studies, human–machine communication asks how people make sense of AI as a communicator (Guzman and Lewis, 2020), and AI-mediated communication asks what happens when AI alters messages between people, with examples such as smart replies and auto-completion (Hancock et al., 2020). Communication and cultural scholarship explains why recognition matters. Jenkins et al. (2006, p. 3) define a participatory culture as one in which "members believe their contributions matter, and feel some degree of social connection with one another." Chalk (2022) shows improvised collective play as cultural production sustained by non-economic values such as participation, enjoyment and community. Neville (2026) shows how technologies are rolled out into practices that outsiders misrecognise as non-technical.

Where this work fits and what it challenges. The project takes from cultural scholarship the stakes of recognition. It questions whether a system's record of contribution can stand in for recognition.

Against HCI work focused on control and decision-making, it asks how authorship and responsibility are negotiated over time, by the people making the work. It also treats the model and the asset library as sources of interpretation, not neutral containers. Documented challenges include stereotyped images from text-to-image systems (Bianchi et al., 2023), value profiles in five GPT models resembling English-speaking and Protestant European survey responses (Tao et al., 2024), and a preference for Western-associated entities among multilingual and Arabic models prompted in Arabic (Naous et al., 2024). A 2022 benchmark found models of that time below human accuracy on paired figurative phrases (Liu et al., 2022). These results concern specific models and tasks, and I will not benchmark models. Instead, I will record how metaphor, idiom and culturally specific phrases are read when participants use them, and how my asset choices privilege particular associations. In professional practice, the project builds on research on improvisation as methodology and socially responsive creative technologies (Caines, 2020; Caines and Watson, 2023). It also builds on recent research-creation on generative AI in teaching, which frames work with AI tools "as a kind of critical improvisation practice" involving "a constant cycle of navigating power, adaptation, surprise, failure, and creation" (Caines, 2026, p. 193). This project asks how that power is negotiated among improvisers when the material is one another's contributions.

Relationship to my master's research. My master's research at Simon Fraser University asks how interfaces make a person's state perceptible to others, through label-free heart-rate cues for remote teammates in a cooperative Unity game and a Unity environment that responds to the player's heart rate. The teammate study is accepted as an extended abstract (poster) at ACM ISS 2026 (Jahandideh et al., 2026), and the second manuscript is under revision following peer review (Jahandideh, 2026). In the teammate study, participants tied their willingness to use the cue to control over it, asking for opt-in sharing, an immediate off switch and separate permissions (Jahandideh et al., 2026). As the designer, I also noticed that the classification behind the display still sorted people into evaluative categories they never saw. Both lead to this dissertation's question, who controls how a shared system interprets people, now asked of deliberate creative contributions rather than bodily signals.

5 Proposed chapter outline

1. Introduction: AI mediation and whose contribution counts.
2. Literature and framework: co-creation, credit, contestation and human–machine reconfiguration.
3. Methodology: research-creation, workshops and reflexivity.
4. The environment as a creative work and the ownership its design assumes.
5. Recognition and authority: how contributions are noticed, transformed, taken up or refused.
6. Authorship: attribution, recognition and responsibility, and where they diverge.

7. Conclusion.

6 Proposed timeline

Dates assume full-time study from Fall 2027 and approval of this proposal in the winter of Year 2; they will be revised with the supervisory committee.

Jan–Apr 2029	Proposal approval; ethics application; system prototype tests with a local model.
May–Aug 2029	Scoping interviews; transcription and initial analysis.
Sep–Dec 2029	Co-design sessions; revision of starting rules and scores.
Jan–Jun 2030	Workshops with three groups; reflection interviews.
Jul–Dec 2030	Episode and interview analysis; documentation of the creative work.
Jan–Jun 2031	Writing and revision of chapters.
Jul–Aug 2031	Submission and defence (Year 5 held in reserve).

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